Code No: 133BD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, February - 2024
MATHEMATICS - IV
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, MIE, MSNT)
Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Define Analytic function. [2]
b) If $w = \log z$, find $\frac{dw}{dz}$ and determine where w is non-analytic. [3]
c) Find the zeros of the function $\sin\left(\frac{1}{z}\right)$. [2]
d) Show that the function e^z has an essential singularity at $z = \infty$. [3]
e) Define conformal mapping. [2]
f) Find the fixed points of the transformation $w = \frac{z-1+i}{z+2}$. [3]
g) State dirichlet conditions. [2]
h) State Fourier integral in complex form. [3]
i) In heat flow, what is meant by transient state. [2]
j) Classify the second order partial differential equations. [3]

PART – B

(50 Marks)

2.a) Show that $f(z) = xy + iy$ is everywhere continuous but it is not analytic.
b) If $f(z) = u + iv$ is an analytic function, show that u(x,y) =C and v(x,y) =K represent two families of orthogonal curves. [5+5]

OR

3.a) Find the conjugate harmonic function of the harmonic function $u = a(1 + cos\theta)$.
b) Show that xy^2 cannot be real part of an analytic function. [5+5]

4.a) Evaluate $\int_c (y - x - 3x^2 i)dz$, where *c* consists of the line segments from *z* = 0 to *z* = *i* and the other from *z* = *i* to *z* = 1+*i*.
b) Expand the function $f(z) = \frac{z}{(z-1)(z-3)}$ as Taylor's series about $0 < |z - 1| < 2$. [5+5]

OR

5. Show that using residue theorem, $\int_0^{2\pi} \frac{1}{a+bsin\theta} d\theta = \frac{2\pi}{\sqrt{a^2-b^2}}, a > b > 0$. [10]

6. Evaluate $\int_0^{2\pi} \frac{sin^2\theta}{a+bcos\theta}\, d\theta$; (a>b>0). [10]

OR

7. Find the bilinear transform which maps the points $z = 0, -i, -1$ into the points $w = i, 1, 0$. Find the image of the line $y = mx$ under this transformation. [10]

8. Find Fourier sine transform of $f(x) = \frac{1}{x(x^2+a^2)}$ and hence deduce cosine transform of $\frac{1}{x^2+a^2}$. [10]

OR

9. If $f(x) = \begin{cases} x, & 0 < x \le \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x < \pi \end{cases}$ then Show that

a) $f(x) = \frac{4}{\pi}\left[sinx - \frac{sin3x}{3^2} + \frac{sin5x}{5^2} - \cdots\right]$

b) $f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left[\frac{cos2x}{1^2} + \frac{cos6x}{3^2} + \frac{cos10x}{5^2} + \cdots\right]$ [5+5]

10. The ends A and B of a rod 20 cm long have the temperatures at 30°C and 80°C until steady-state prevails. The temperatures of the ends are changed to 40°C and 60°C respectively. Find the temperature distribution in the rod at time t. [10]

OR

11. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially in a position given by $y = y_0 \sin^2(\pi x/l)$. If it is released from rest from this position, find the displacement $y(x, t)$. [10]

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